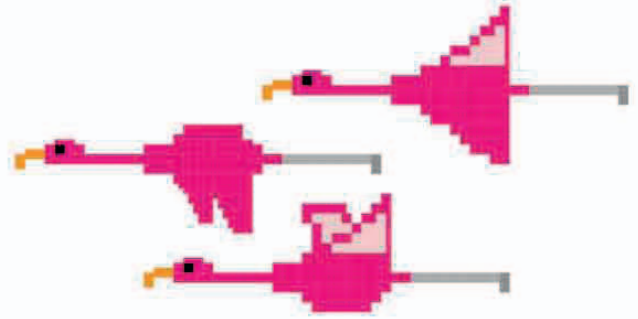


Numbering the characters

Each character in a string is allocated a number according to its position. This position number can be used to look at individual letters or symbols, or to pull them out of a string.



1 Count begins from zero

When counting the positions, Python starts at 0. The second character is in position 1, the third in position 2, and so on.

```
>>> a = 'FLAMINGO'
```



The sixth letter, "N", is in position 5

0 1 2 3 4 5 6 7

The first character, "F", is in position 0

2 Counting the characters

The position number is called an "index". It can be used to pull out a particular letter from a string.

```
>>> a[3]
```

Square brackets go around the index

'M'

The character in position 3 from the variable "a"

The last character, "O", is in position 7

3 "Slicing"

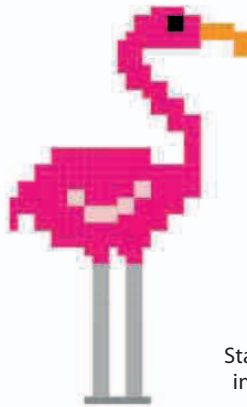
Two indexes can be used to pull out a part of the string or "slice" it. The letter in the last position isn't included.

```
>>> a[1:7]
```

Colon defines the range of characters

'LAMING'

A slice from index 1 to index 6 of variable "a"



4 From the start or the end

If you leave off the start or end index, Python will automatically use the first or the last character of the string.

```
>>> a[:3]
```

'FLA'

```
>>> a[3:]
```

'MINGO'

Ends at index 7

Starts at index 0

Apostrophes

Strings can go in single or double quotation marks. However, the string should start and end with the same type of quotation mark. This book uses single quotes. But what happens if you want to use an apostrophe in your string?

```
>>> print('It\'s a cloudy day.')
It's a cloudy day.
```

The apostrophe is included in the string

△ Escaping the apostrophe

So Python doesn't read an apostrophe as the end of the string, type a "\" before it. This is called "escaping" it.